

CryoStack

A Unified Cyberinfrastructure Platform for Reproducible Ice-Sheet Modeling, Data Assimilation, and Scientific Computing across HPC and Cloud Environments

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The challenge is bigger than the model



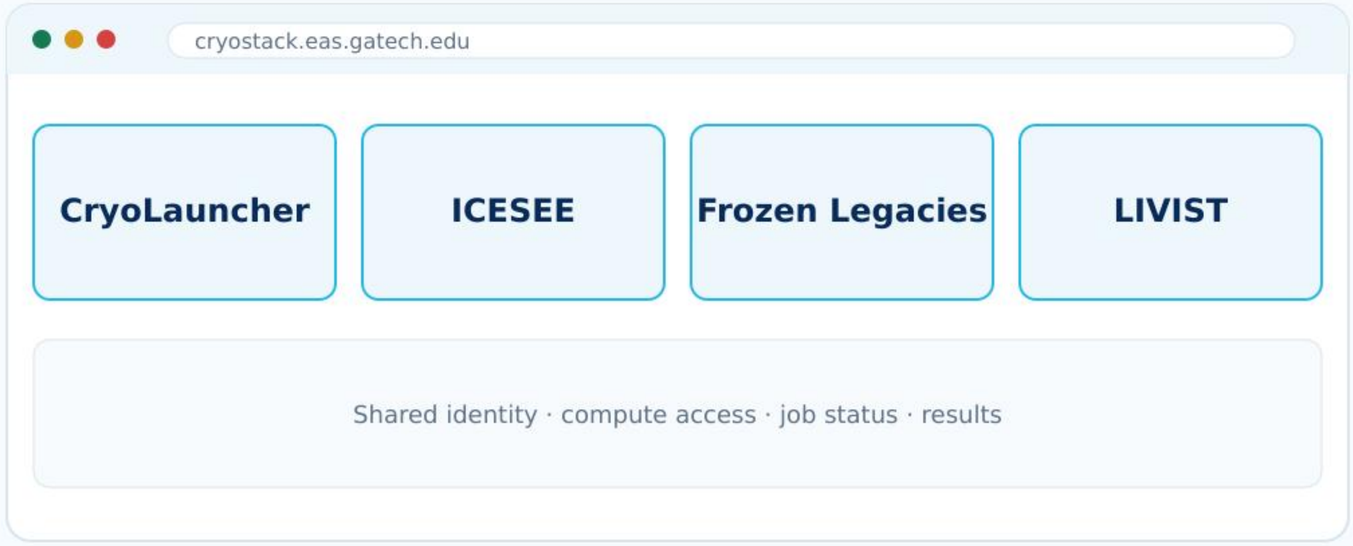


Every one of those steps is a place where reproducibility can quietly break.

Introducing CryoStack

Science first. Infrastructure underneath.

Configure · Launch · Monitor · Visualize · Reproduce



→

HPC / Cloud

Your compute

One platform. Different scientific workflows.



CryoLauncher

ISSM · Icepack

Model configuration and execution



ICESEE

Ensemble data assimilation

State and parameter estimation



Frozen Legacies

Historical Antarctic radar

Cataloging · interpretation · tracing



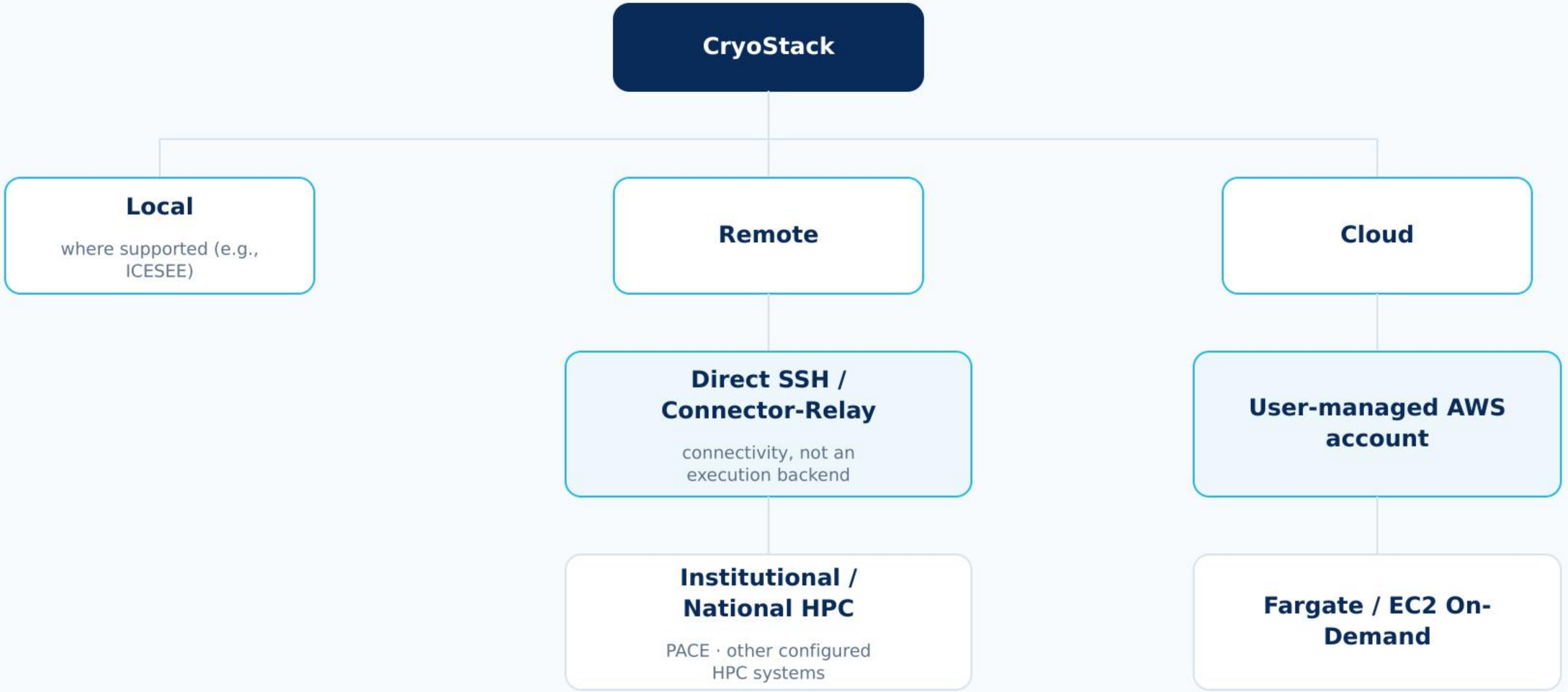
LIVIST

Englacial temperature

Radar · borehole observations

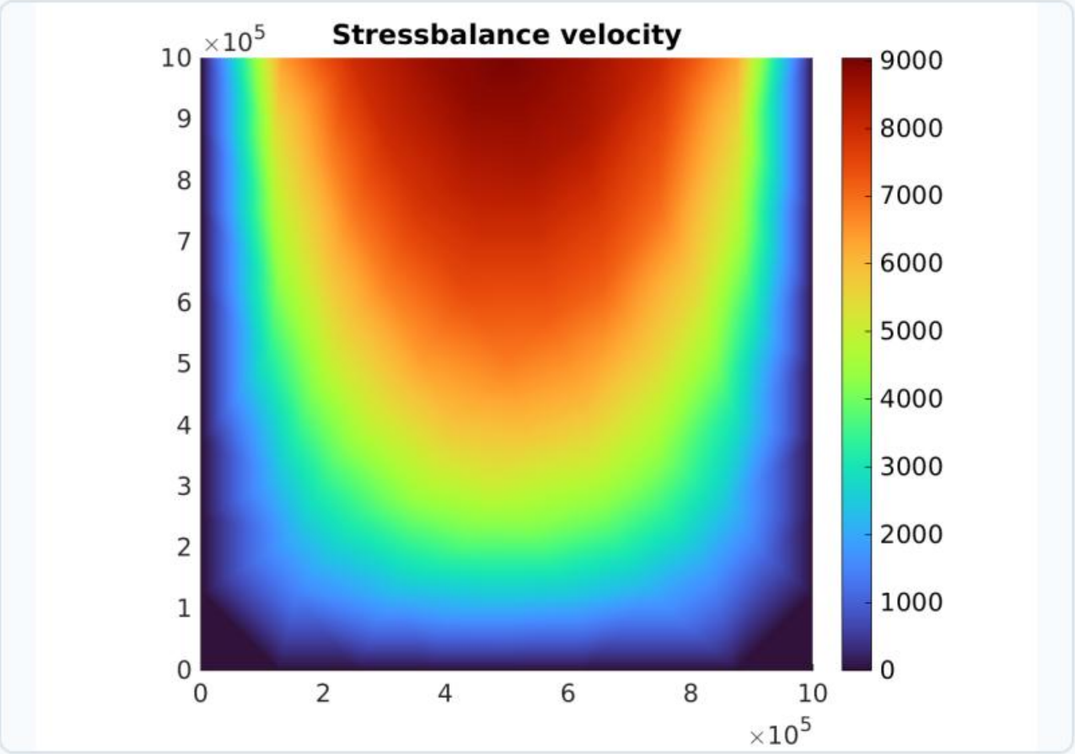
Shared cyberinfrastructure. Application-specific science.

From your browser to the compute you already use

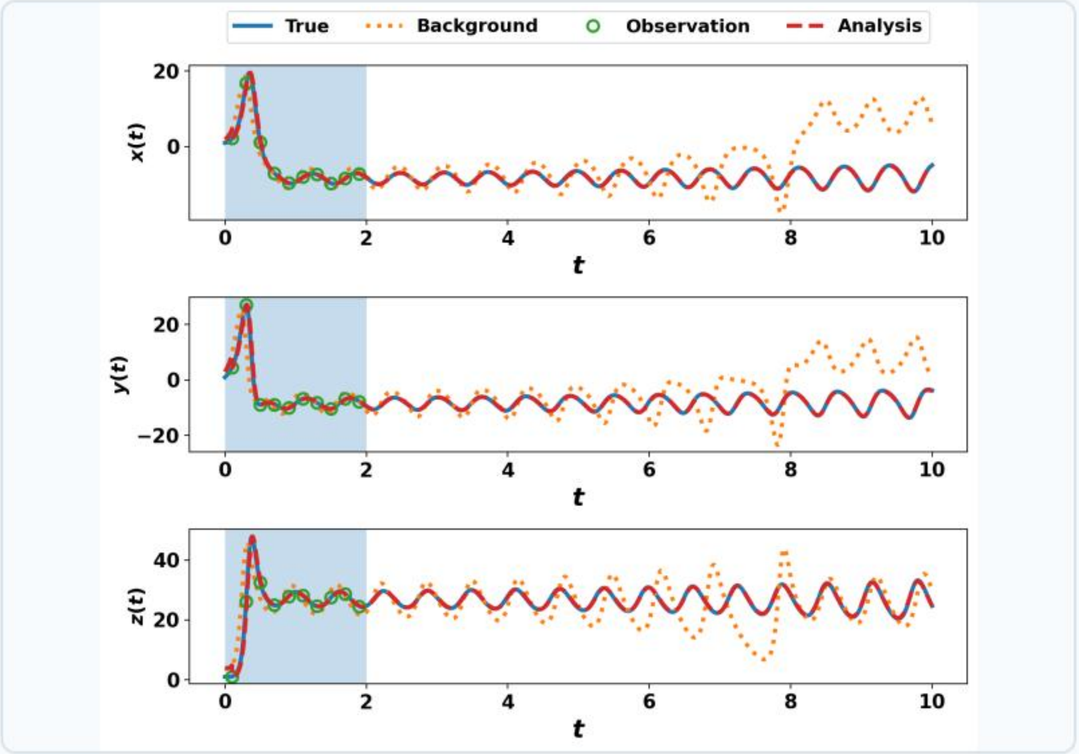


CryoLauncher supports Remote and Cloud. Local applies where supported, particularly ICESEE.

It is already running real science



ISSM -- Remote execution (PACE via Connector)



ICESEE -- ensemble data assimilation

Also demonstrated on AWS: ISSM & Icepack -- Fargate / EC2 On-Demand

The experiment is more than its final output



CryoStack is built to carry an experiment's context through its lifecycle -- not just the final result.

CryoStack is live.

Explore the platform. Run a workflow. Help us build what comes next.

cryostack.eas.gatech.edu

Documentation · cryostack.eas.gatech.edu/documentation.html

SCAN TO EXPLORE

More models · datasets · analysis tools · community workflows -- next: PISM, Elmer/Ice, SICOPOLIS

See CryoStack in action

From configuration to scientific results



ISSM — Remote HPC



ISSM — Cloud



Icepack — Cloud

Short excerpts here -- full recordings will be published as CryoStack tutorials.